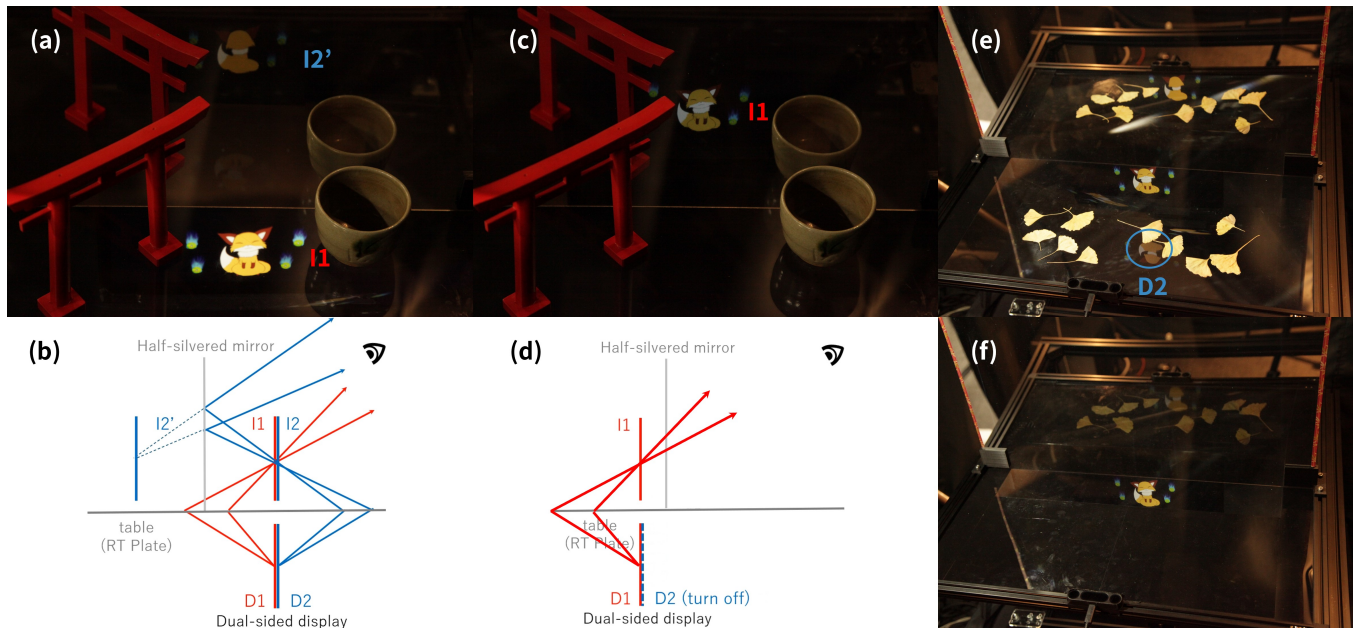


# MMM: Mid-air image Moving in and out of the Mirror with backward glance in the mirror

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**Figure 1:** (a) The appearance and (b) optical system when the mid-air image is outside the mirror. (c) The appearance and (d) optical system when the mid-air image is inside the mirror. (e) without VCF (f) with VCF

## Abstract

We propose an optical system displaying mid-air images that move between the inside and outside of a mirror with backward glance in the mirror. Previous research has not been able to achieve all three of these goals: the standing image floats in space, moves across the mirror's surface, and is reflected in the mirror. Proposed system "MMM" achieves all three of these goals and presents a worldview of the mirror as an adjacent virtual world. A magic mirror application that brings characters into the real world from a display have been developed. In this application, the mirror space connects the cyberspace inside the display with the real world.

## CCS Concepts

• **Human-centered computing** → **Mixed / augmented reality; Displays and imagers.**

## Keywords

mid-air image, mixed reality, mirror display

## ACM Reference Format:

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## 1 Introduction

A mirror merely reflects real space, but the mirror world has often sparked the imagination as an otherworldly realm, as in "Through the Looking-Glass, and What Alice Found There." [Carroll 2010] The real augmentation through the optical phenomenon of mid-air images provides users with the visual experience of physical phenomena that would not normally occur without wearing any device. We propose an optical system displaying mid-air images that move between the inside and outside of a mirror with backward

glance in the mirror. Previous research has not been able to achieve all three of these goals: the standing image floats in space [Kakehi and Naemura 2005], moves across the mirror's surface [Kato and Naemura 2006], and is reflected in the mirror [Sano et al. 2023]. Proposed system “MMM” achieves all three of these goals and presents a worldview of the mirror as an adjacent virtual world.

## 2 Proposed Method

The proposed system consists of an Retro-Transmissive Plate (RT Plate) [Otsubo 2014] used as the table surface, a half-silvered mirror on the table, and a dual-sided display moving under the RT Plate, as shown in Fig. 1 (b) and (d).

### 2.1 Moving mid-air image in and out of mirror

The RT Plate emits incident light in a direction symmetric to its plane. This causes the diffused light from the light source to refocus at a point symmetrical to the plate, thereby forming an mid-air image. Thus, the surfaces D1 and D2 of the dual-sided display form mid-air images I1 and I2, respectively, directly above the display.

When the mid-air image is outside of the mirror, the optical path of I1, indicated in red in the figure, is directly visible to the eye. In contrast, the optical path of I2, indicated in blue in the figure, does not enter the eye directly but is reflected by a half-silvered mirror, resulting in the mid-air image appearing at positions I1 and I2', respectively, as shown in Fig. 1 (a). D2 is turned off when the mid-air image is inside the mirror as shown in Fig. 1 (d).

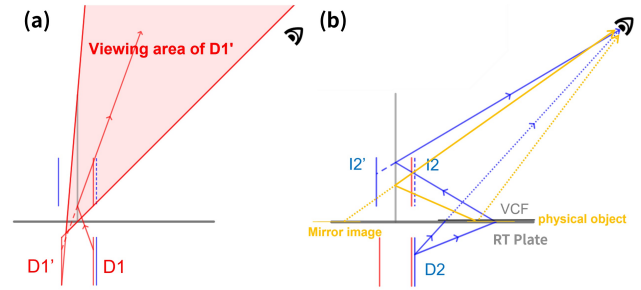
Because both mid-air images are generated using a single dual-sided display as the light source, moving the display allows the two mid-air images to move around while maintaining a mirror-conjugate relationship. The light source can be positioned in front of the mirror (Fig. 1 (b)) or behind it (Fig. 1 (d)), enabling the mid-air images to transition continuously between outside (Fig. 1 (a)) and inside (Fig. 1 (c)) the mirror as the light source moves.

### 2.2 Preventing stray light

When the mid-air image is outside the mirror, D1 reflects on the mirror surface, and D2 is turned on, causing stray light from their transmitted beams.

D1', the mirrored image of D1, can only be seen from the viewing position where D1' and the half-silvered mirror overlap. By limiting the forward movement range of the display, the viewing position can be kept outside the viewing area of D1', as shown in Fig. 2 (a). D2 can be reduced by placing a View Control Film (VCF) on the RT Plate in areas where I1 does not reach similarly to ReQTable [Takenawa et al. 2022]. VCF is an element that transmits light in one direction while blocking it in another. VCF blocks the transmitted light from D2, making it invisible, while the imaging light that forms I2 passes through, allowing I2' to remain visible. As shown in Fig. 2 (b).

VCF darkens I2', but it also allows for content expansion. If an object is placed between the VCF and the RT Plate, the object will appear to exist only inside the mirror and not outside of it. The fact that not only the mid-air image but also physical objects appear only inside the mirror helps to present the mirror as an adjacent virtual world. This is a new approach that not only reduces stray light but also expands content. Without VCF, the transmitted light from D2 and the physical leaves in front of the mirror are visible,



**Figure 2: Prevent Stray light.** (a) viewing position is outside the viewing area of D1'. (b) D2 is shielded by VCF. VCF also conceals physical objects, making them visible only inside the mirror.

as shown in Fig. 1 (e). With VCF, the transmitted light from D2 becomes invisible, and the leaves appear only inside the mirror and not in front of it, even though they are actually only in front of the mirror, as shown in Fig. 1 (f).

## 3 Application

In this paper, we explain a magic mirror that brings characters into the real world from display. When the display is reflected in the mirror, the mirrored character within the display leaps out of the mirror and into the real space outside the mirror. And touch interactions with the character in real space using a hand tracker. In this application, the mirror space functions as a virtual world adjacent to reality, connecting the cyberspace inside the display with the real world.

## 4 Conclusion

We proposed “MMM”, an optical system displaying mid-air images that move between the inside and outside of a mirror with backward glance in the mirror. In addition to reducing stray light, we made it possible for physical objects to appear only inside the mirror. We developed a magic mirror application that bring out the system's characteristics and presents a worldview of the mirror as an adjacent virtual world.

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